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Project Name	CRM Application for jewel management
Maximum Marks	

CHAPTER-4

PROJECT DESIGN

4.1 - Existing System (Problem Solution Fit) :

Salesforce is a powerful cloud-based CRM platform widely adopted across industries for managing customer relationships, sales, and operations. In jewel management, it offers flexibility, scalability, and integration capabilities—but it's not without its challenges.

⚠ Problem Areas in Salesforce Implementation

Challenge	Description
Complex Customization	Jewel businesses often need highly tailored workflows, which can be hard to configure.
Integration Difficulties	Connecting Salesforce with legacy systems (e.g., POS, ERP) can be technically demanding.
Data Quality Issues	Migrating and maintaining clean customer and inventory data is a common hurdle.
User Adoption	Staff unfamiliar with CRM tools may resist or misuse the system.
Performance Bottlenecks	High data volumes (e.g., product catalogues, customer records) can slow down the system.
Cost & Licensing	Salesforce's pricing model may be steep for small jewel businesses.

✓ Solution Fit: Why Salesforce Still Works

Despite the challenges, Salesforce remains a strong fit when implemented strategically:

- **Custom Objects & Flows:** Tailor data models for jewellery items, repairs, and customer preferences
- **AppExchange Ecosystem:** Access third-party apps for inventory, billing, and loyalty programs
- **Automation Tools:** Use Flows and Process Builder to streamline repetitive tasks
- **Analytics & Dashboards:** Real-time insights into sales, stock levels, and customer engagement
- **Mobile Access:** Enable jewellers to manage operations on the go
- **Security & Compliance:** Robust data protection for sensitive customer and transaction data

4.2 - Proposed System (Proposed Solution) :

The proposed solution leverages **Salesforce’s cloud-based CRM platform** to create a scalable, secure, and customizable system tailored to the unique needs of jewellery businesses. It addresses the limitations of traditional systems and enhances customer engagement, inventory control, and operational efficiency.

🔑 Key Components of the Proposed Solution

Module	Functionality
Customer Management	Centralized profiles, purchase history, loyalty tracking
Inventory Control	Real-time stock updates, barcode/RFID integration, alerts for low/high demand
Sales & Billing	Automated invoicing, discount handling, POS integration
Order & Repair Tracking	Custom orders, repair status updates, notifications
Marketing Automation	Email/SMS campaigns, personalized offers, lead nurturing
Analytics & Reporting	Dashboards for sales trends, customer behaviour, inventory turnover

⚙️ Salesforce Features Utilized

- **Custom Objects & Fields:** For jewellery items, repairs, and customer preferences
- **Flows & Process Builder:** Automate tasks like follow-ups, order status updates
- **Reports & Dashboards:** Visualize KPIs and business metrics
- **AppExchange Integrations:** Connect with accounting, e-commerce, and loyalty apps
- **Mobile App Access:** Manage operations on-the-go
- **Security & Compliance:** Role-based access, data encryption, audit trails

🚀 Benefits of the Proposed System

- **Personalized Customer Experience:** Tailored recommendations and loyalty rewards
- **Operational Efficiency:** Reduced manual work and errors
- **Scalability:** Easily adapts to business growth and new product lines
- **Data-Driven Decisions:** Actionable insights from real-time analytics
- **Enhanced Collaboration:** Unified platform for sales, support, and inventory teams

4.3 - Solution Architecture :

This architecture is designed to support core business functions like customer engagement, inventory tracking, sales automation, and analytics—all within the Salesforce ecosystem.

🔧 Core Layers

Layer	Components
Presentation Layer	Salesforce Lightning UI, Mobile App, Customer Portal
Application Layer	Apex Classes, Lightning Web Components (LWC), Flows, Process Builder
Data Layer	Standard & Custom Objects (e.g., Jewel_Item__c, Repair_Order__c, Customer__c)
Integration Layer	REST/SOAP APIs, AppExchange Connectors (POS, Accounting, E-commerce)
Security Layer	Role-based access, Field-level security, OAuth 2.0, Audit Trails

🔄 Data Flow & Interactions

- **Customer Portal:** Customers browse products, place orders, and track repairs
- **Salesforce UI:** Staff manage inventory, customer profiles, and sales workflows
- **Automation:** Flows trigger follow-ups, stock alerts, and loyalty rewards
- **Reports & Dashboards:** Real-time insights into sales, inventory, and customer engagement
- **External Systems:** Sync with POS, accounting tools (e.g., Tally), and e-commerce platforms

📦 Technology Stack

Component	Technology
CRM Platform	Salesforce Sales Cloud
Custom Development	Apex, LWC, Visualforce
Integration	MuleSoft, REST APIs, AppExchange Apps
Analytics	Salesforce Reports & Dashboards
Mobile Access	Salesforce Mobile App

🚀 Deployment Strategy

- **Sandbox Testing:** Validate workflows and integrations
- **Change Sets:** Deploy metadata between environments
- **Monitoring:** Use Salesforce Shield for audit and performance tracking
- **Training:** On board staff with Trailhead modules and custom guides